Kernix 0.7.1 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

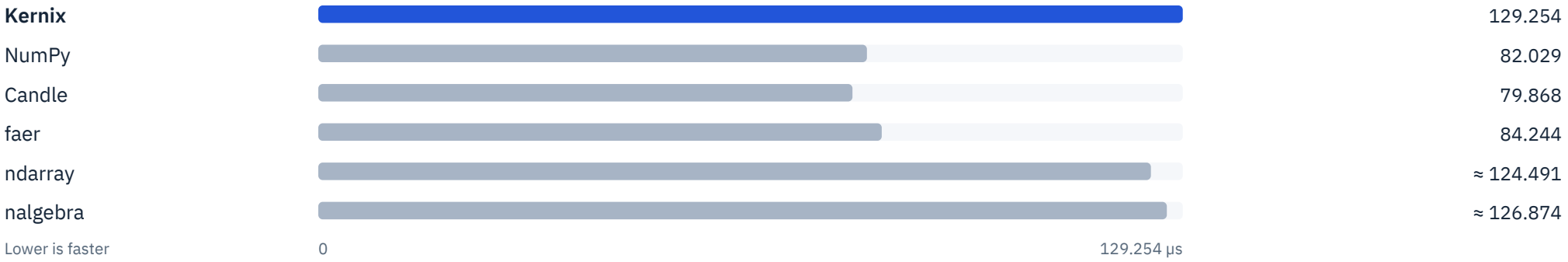
14 workloads

50 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

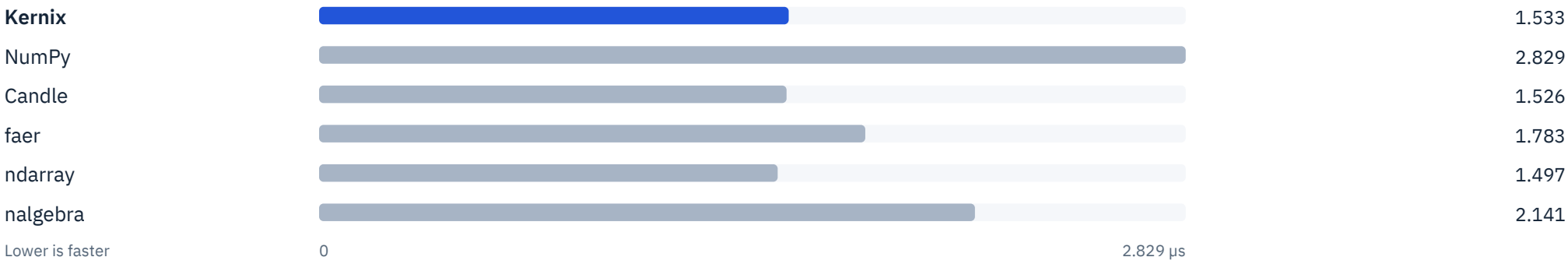
Kernix 0.7.1 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

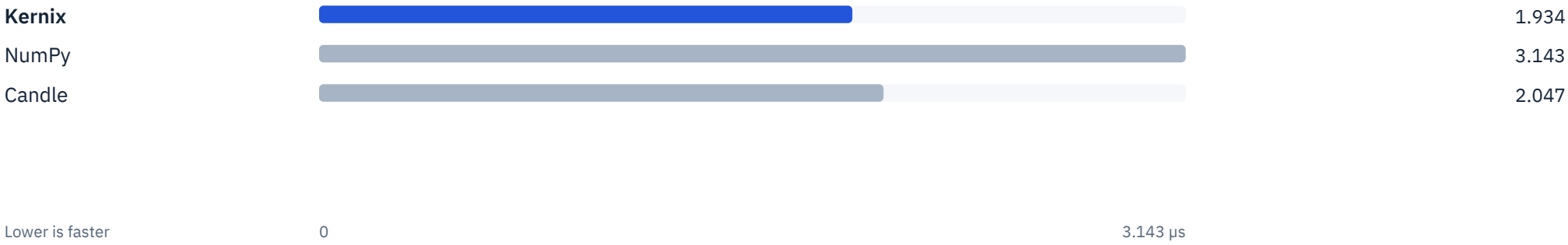
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

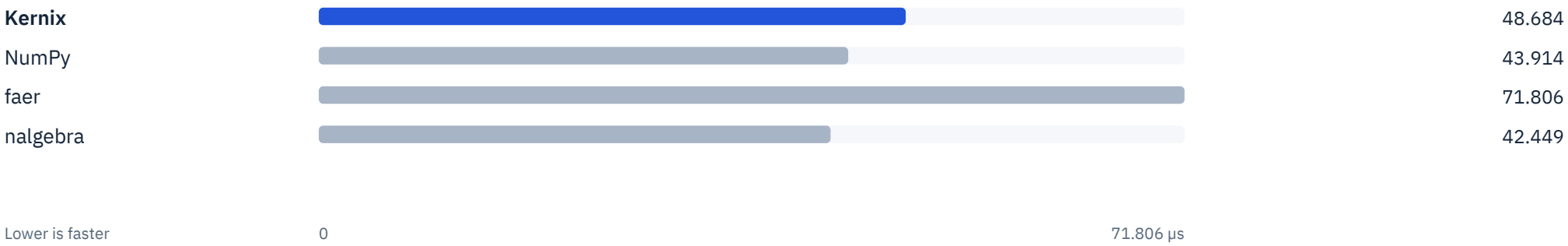


Kernix 0.7.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output

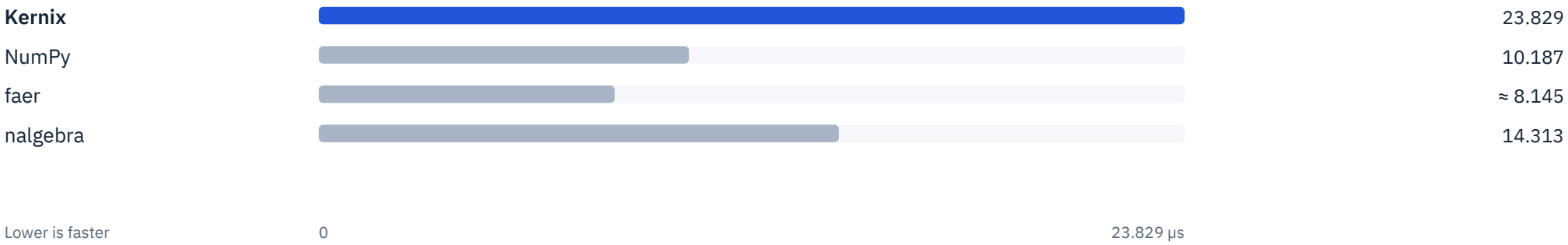


Kernix 0.7.1 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.7.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

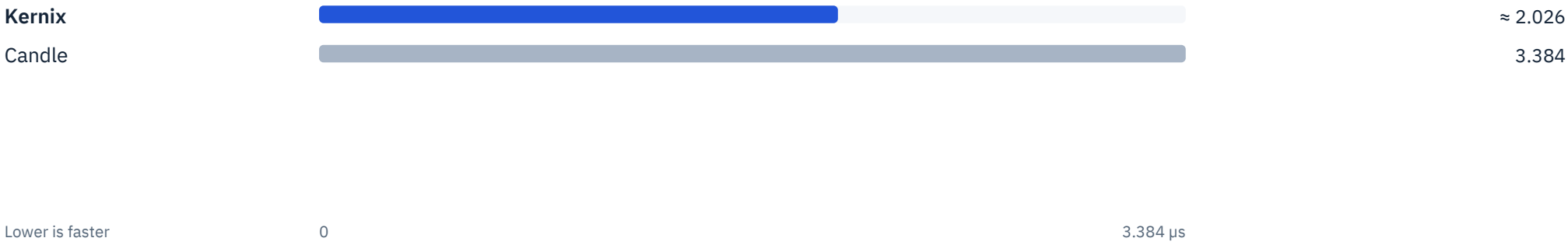
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.7.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

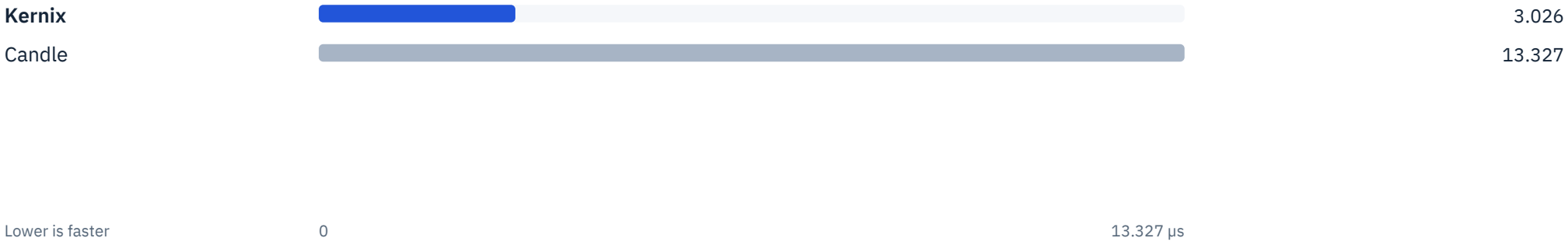
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.7.1 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

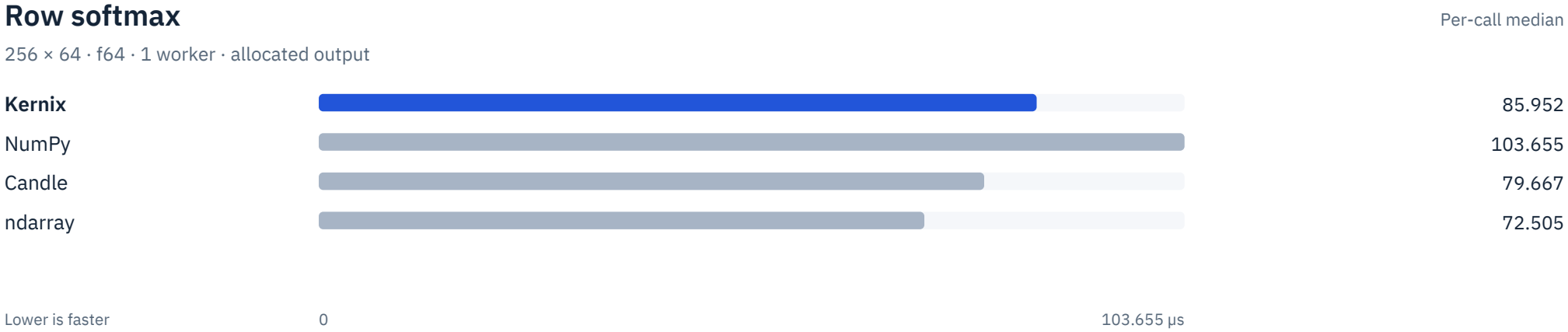


Kernix 0.7.1 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



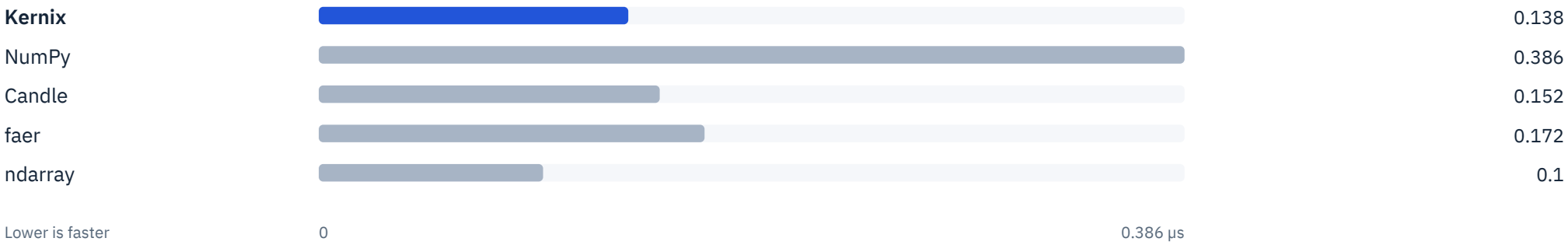
Kernix 0.7.1 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1533	1.533
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1497	1.497
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2141	2.141
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1783	1.783
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1526	1.526
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2829	2.829
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	129254	129.254
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	124491.00000000001	124.49100000000001
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	126873.99999999999	126.87399999999999
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	84244	84.244

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	79868	79.868
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	82029	82.029
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1934	1.934
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2047	2.047
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3143	3.143
Thin QR / Kernix 96 × 32 · f64	48684	48.684
Thin QR / nalgebra 96 × 32 · f64	42449	42.449
Thin QR / faer 96 × 32 · f64	71806	71.806
Thin QR / NumPy 96 × 32 · f64	43914	43.914
Thin QR / Kernix 256 × 64 · f64	596452	596.452

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / nalgebra 256 × 64 · f64	438001	438.001
Thin QR / faer 256 × 64 · f64	475510	475.51
Thin QR / NumPy 256 × 64 · f64	399693	399.693
Cholesky factorization / Kernix 64 × 64 · f64	23829	23.829
Cholesky factorization / nalgebra 64 × 64 · f64	14313	14.313
Cholesky factorization / faer 64 × 64 · f64	8145.000000000001	8.145000000000001
Cholesky factorization / NumPy 64 × 64 · f64	10187	10.187
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2413	2.413
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4036.9999999999995	4.0369999999999995
Model inference / Kernix batched normalized/sliced MLP · f64	2026.0000000000002	2.0260000000000002

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	3384	3.384
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2525	2.525
Model recording + forward / Candle batched normalized/sliced MLP · f64	4984	4.984
Model backward / Kernix prepared graph → fresh gradients · f64	3026	3.026
Model backward / Candle prepared graph → fresh gradients · f64	13327	13.327
Parameter update / Kernix transactional SGD, fixed gradients · f64	536	0.536
Row softmax / Kernix 256 × 64 · f64	85952	85.952
Row softmax / ndarray 256 × 64 · f64	72505	72.505
Row softmax / Candle 256 × 64 · f64	79667	79.667
Row softmax / NumPy 256 × 64 · f64	103655	103.655

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	138	0.138
Create zeros / ndarray 64 × 64 · f32	100	0.1
Create zeros / faer 64 × 64 · f32	172	0.172
Create zeros / Candle 64 × 64 · f32	152	0.152
Create zeros / NumPy 64 × 64 · f32	386	0.386
Create ones / Kernix 64 × 64 · f32	202	0.202
Create ones / ndarray 64 × 64 · f32	154	0.154
Create ones / faer 64 × 64 · f32	196	0.196
Create ones / Candle 64 × 64 · f32	236	0.236
Create ones / NumPy 64 × 64 · f32	2441	2.441